CSSD 111: Introduction to Computer Systems

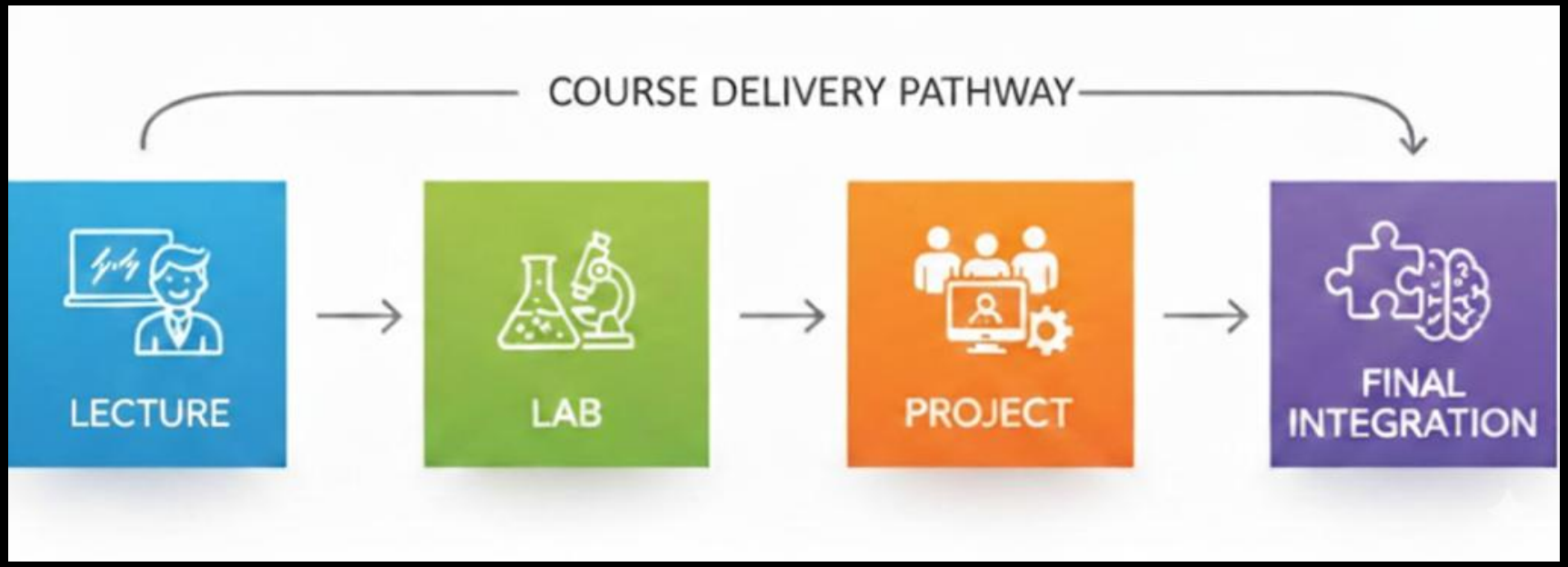
Week 1: Introduction to the World of Technology

- Lecturer: Mutala Mohammed
- Department of Computer Science
- Faculty of Computing and Information Systems

- Academic Year: 2025/2026.

COURSE OVERVIEW (BRIEF REMINDER)

- Course structure (2h Lecture + 1h Lab)
- Cumulative Enterprise Project introduction
- Assessment structure overview



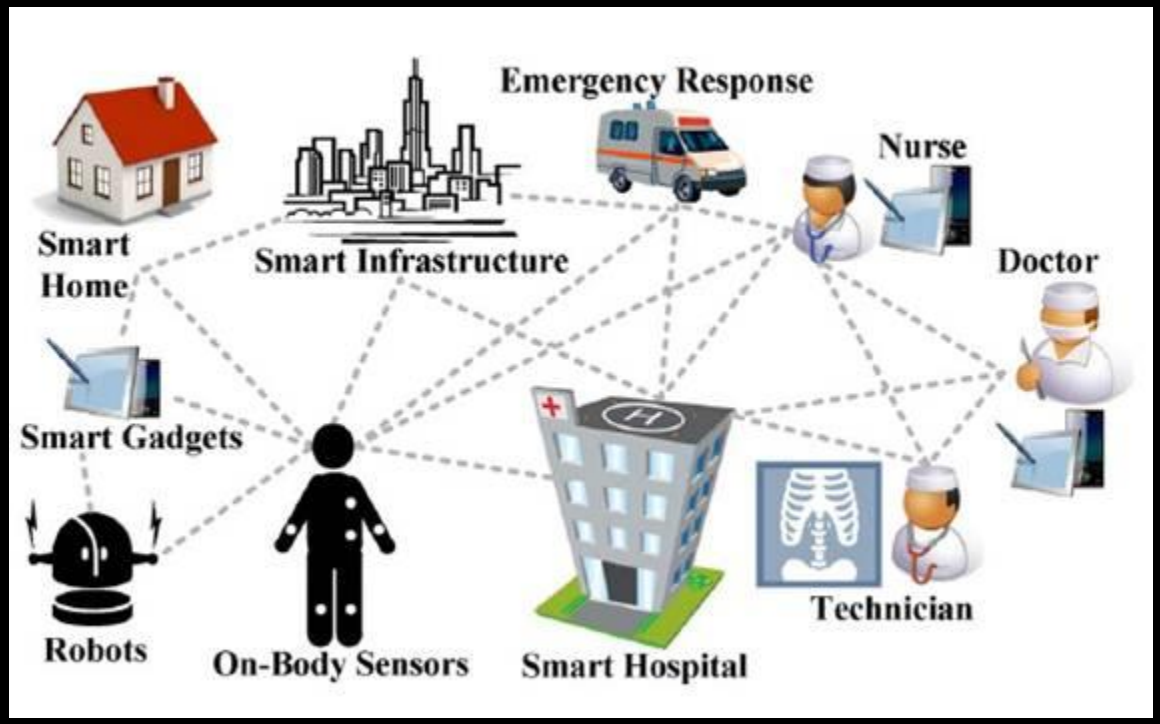
LEARNING OBJECTIVES

By the end of this week, students should be able to:

- Why study Computer systems?
- Define data, information, knowledge, system
- Explain what a computer system is
- Identify major categories of computing devices
- Distinguish personal computing from enterprise computing
- Organize digital files professionally for lab and integrated project work.

• WHY STUDY/IMPACT COMPUTER SYSTEMS?

- The transition from "computer literacy" to "digital fluency"
- Ubiquitous Computing (now Computers are Everywhere)
- Impact of computer on almost all Professional Careers
- Efficient Data Management.



THE UBIQUITY OF COMPUTERS

Ubiquitous Computing:

- Computers are everywhere—in our pockets, cars, and appliances.

The Shift (in our case):

- We are moving from "computer literacy" (knowing how to use a PC) to "digital fluency" (using technology to solve organizational problems).

Application:

- Specific major professions (Marketing, Finance, HR) require we study its own specific systems.

• WHY STUDY/IMPACT COMPUTER SYSTEMS?

Area	Impact
Productivity	Systems automate repetitive tasks, allowing humans to focus on creative problem-solving.
Communication	Global connectivity allows for real-time collaboration across different time zones.
Decision Making	Real-time data processing provides managers with accurate insights to make better business choices.
Security	Understanding systems is the first step in protecting personal and corporate data from cyber threats.

SO, IT'S NOW A MUST WE STUDY COMPUTER SYSTEMS

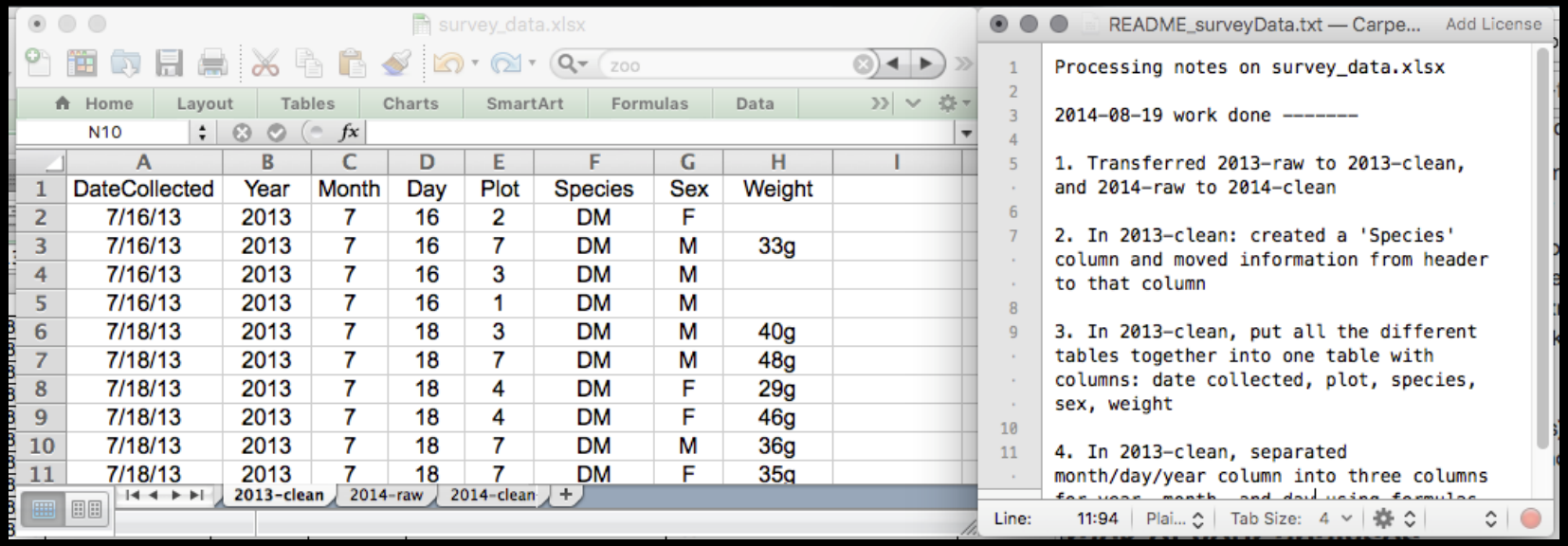
FROM DATA TO INFORMATION TO KNOWLEDGE

What is Data?

Definition: Data = Raw, unprocessed facts.

Examples:

- 25 old
- "Accra"
- 23°C
- 0244018812



	A	B	C	D	E	F	G	H
1	DateCollected	Year	Month	Day	Plot	Species	Sex	Weight
2	7/16/13	2013	7	16	2	DM	F	
3	7/16/13	2013	7	16	7	DM	M	33g
4	7/16/13	2013	7	16	3	DM	M	
5	7/16/13	2013	7	16	1	DM	M	
6	7/18/13	2013	7	18	3	DM	M	40g
7	7/18/13	2013	7	18	7	DM	M	48g
8	7/18/13	2013	7	18	4	DM	F	29g
9	7/18/13	2013	7	18	4	DM	F	46g
10	7/18/13	2013	7	18	7	DM	M	36g
11	7/18/13	2013	7	18	7	DM	F	35g

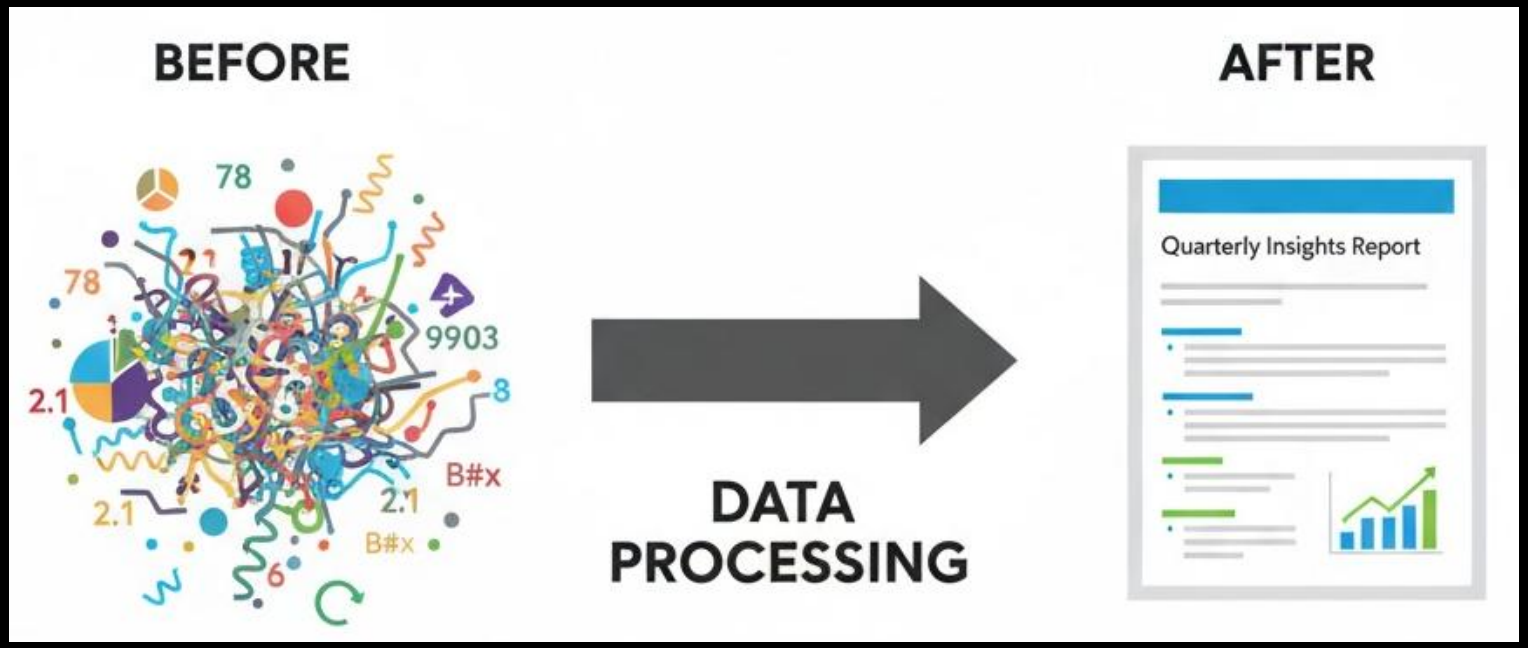
FROM DATA TO INFORMATION TO KNOWLEDGE

What is **Information**?

- Information = Processed data that has meaning.

Example:

- "Total sales this month: GHS 45,000"
- "Temperature is 23°C — cool weather."



FROM DATA TO INFORMATION TO KNOWLEDGE

Data vs Information (Comparison)

Data	**Information**
Raw	Processed
No meaning alone	Meaningful
Input	Output

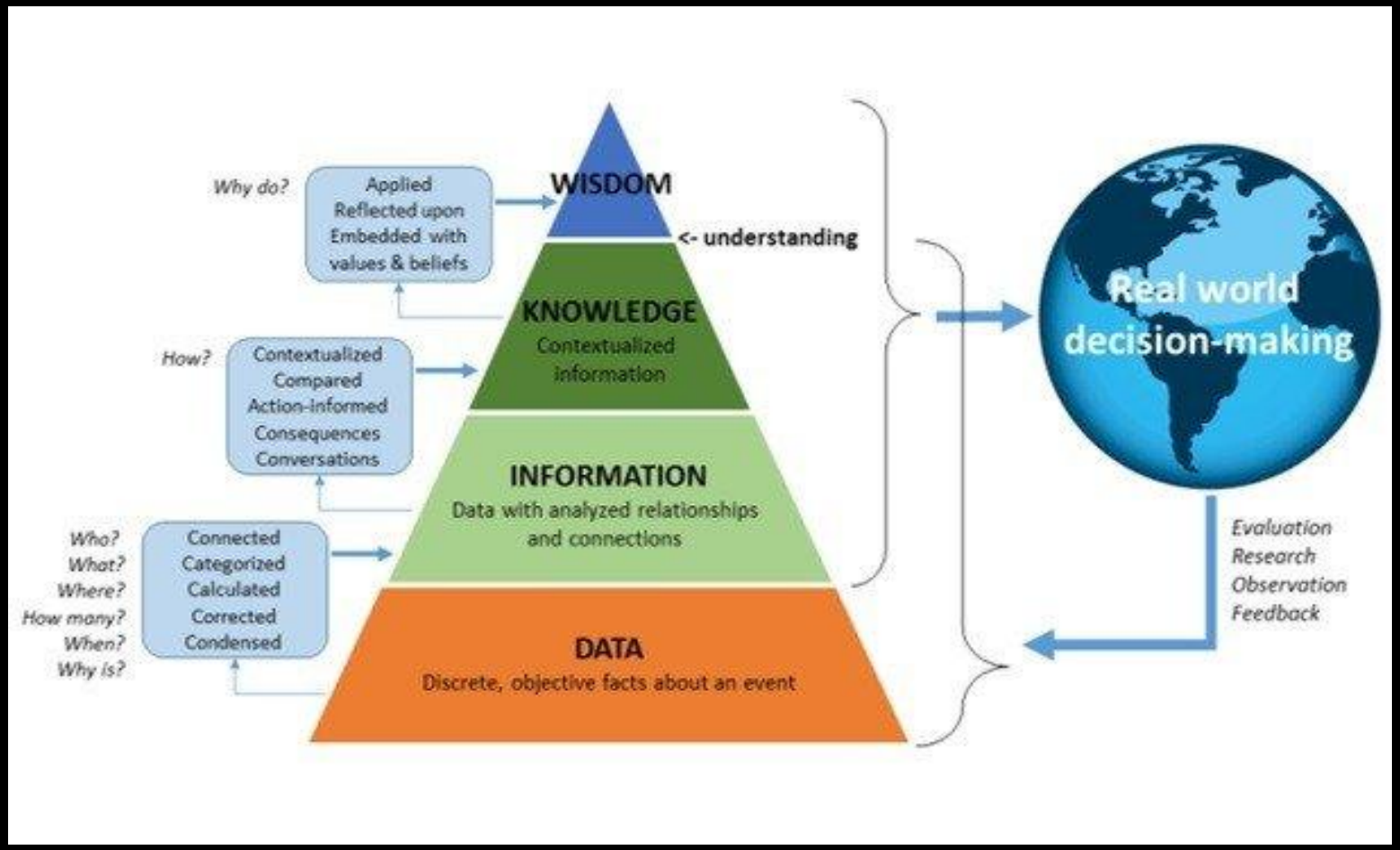
FROM DATA TO INFORMATION TO KNOWLEDGE

- DIKW Pyramid (Data, Information, **Knowledge**, Wisdom)
- Definition:
  - Knowledge (The Application): The ability to use information to make decisions, solve problems, or create strategies.
  - It is information combined with experience and understanding.

Level	Description	Example in a Business Context
Data	Raw facts	"GHS 500"
Information	Data with context	"The price of the component is GHS 500."
Knowledge	Information + Application	"GHS 500 is too expensive compared to our budget; we should find a new supplier."

FROM DATA TO INFORMATION TO KNOWLEDGE

Why: In the "Knowledge Age," the value is in using the **Knowledge** (e.g. car: the engine knowledge + experience by driver) to make strategic decisions.



INFORMATION SYSTEMS (THE "SYSTEM" PART)

- What then is a System?
- *Back in the SHS: circulatory system, digestive system, etc.*
- A System:
  - as a combination of different components/parts/organs/people that work together to produce (to produce) a goal/output
  - The components play different roles,
  - but work together to achieve/towards a main goal.
- What is information systems?
  - Information System is a combination of **Hardware, Software, Data, People, and Procedures** that work together to produce information.
    - Impact: This connects the "Computer" (hardware/software) to the "Business" (people/procedures), which is the core of **Digital Fluency**.

FOUNDATIONS OF THE CONNECTED WORLD

- How these systems talk to each other?
  - Networks
  - For the global system, it's the Internet/Web
- What is a Computer Network?
  - A collection of computers and devices
  - connected together so
  - users can
  - share hardware, software, and data, as well as communicate.

FOUNDATIONS OF THE CONNECTED WORLD

- The Internet vs. The World Wide Web (WWW)
- The Internet:
  - The physical network of networks, linking billions of devices globally via a backbone infrastructure.
- The World Wide Web:
  - One specific resource available via the Internet;
    - it is the collection of Web pages stored on Web servers.

FOUNDATIONS OF THE CONNECTED WORLD

- Accessing the Network:

1. **Hardware**: Network Adapter (built-in) or a Modem, etc.
2. **Gateway**: Internet Service Provider (ISP), e.g. in Ghana.
3. **Authentication**: Logon information (Username and Password) for security.

FOUNDATIONS OF THE CONNECTED WORLD

Related Online terminologies:

- Surfing and Searching the Web
  - Surfing
  - Web Browser
  - Hyperlinks
  - The Home Page
  - Searching
  - Search Sites
  - Search Engines
- Electronic Mail (E-Mail)
  - Email/Web Mail
  - Anonymity & Privacy
- Digital Ethics
  - Citing Sources
  - Plagiarism

WHAT IS A COMPUTER?

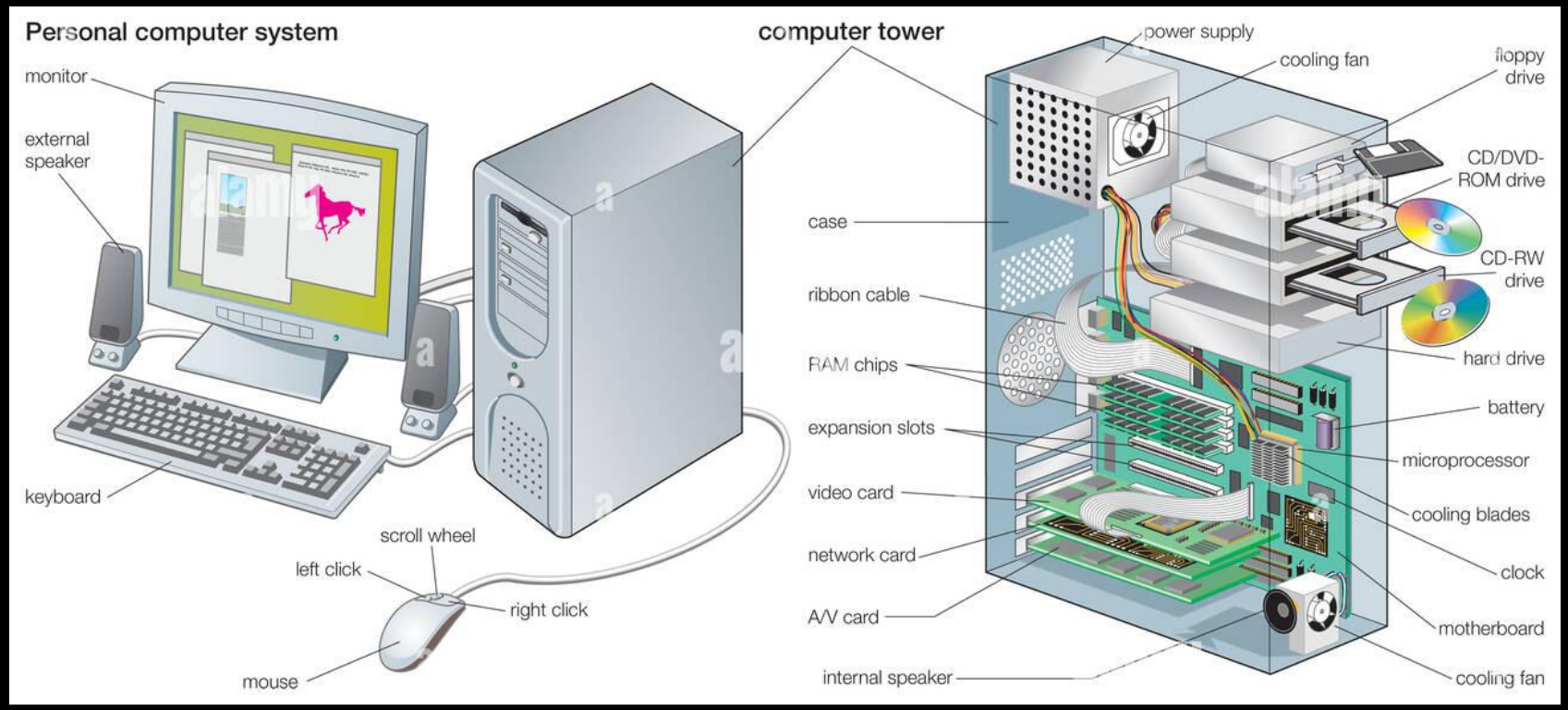
All these systems depend on the computer

Definition:

- A programmable, electronic device that accepts data, performs operations, produces and stores results/output.
- Which is usually abbreviated as:
  - IPO Cycle (Input → Process → Output)
- The process uses: Hardware vs. Software
  - Hardware is the physical "body"; Software is the "mind" or instructions.

BASIC COMPUTER COMPONENTS

- Input Devices
- Processing Unit (CPU)
- Memory & Storage
- Output Devices



TYPES OF COMPUTING DEVICES

- Desktop
- Laptop
- Tablet
- Smartphone
- Server
- Embedded Systems (ATM, POS, Car systems)

THE DIGITAL REVOLUTION

Discuss:

- From paper-based to digital
- Automation of business
- Cloud computing
- Mobile-first economy

PERSONAL VS. ENTERPRISE COMPUTING

Personal Computing: focuses on individual productivity, using:

- Personal laptop
- personal apps
- Social media
- Assignments

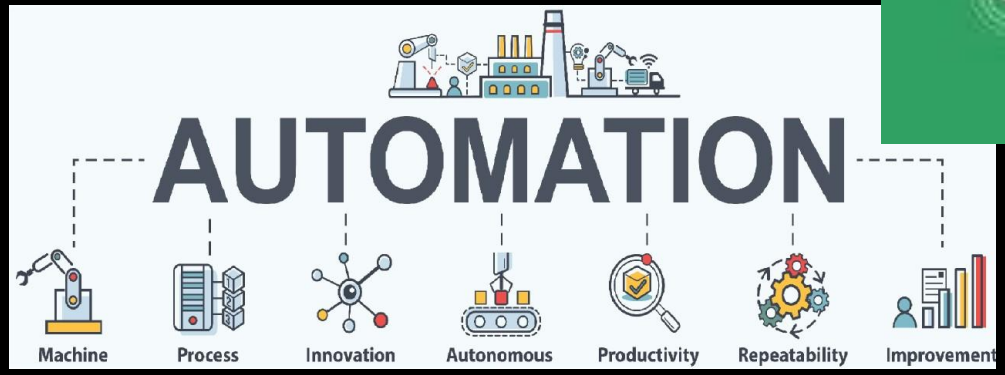
Enterprise Computing: focuses on the whole organization, using:

- Servers
- networked systems
- Databases (like ERP or CRM)
- Security systems

- These connect us to our **Cumulative Enterprise Project**

BENEFITS OF TECHNOLOGY

- Speed
- Automation
- Global communication
- Data analysis



RISKS OF TECHNOLOGY

- Cybercrime
- Data breaches
- Privacy issues
- Misinformation

BBC

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Ghana arrests Nigerians accused of running cyber-crime networks

19 January 2026

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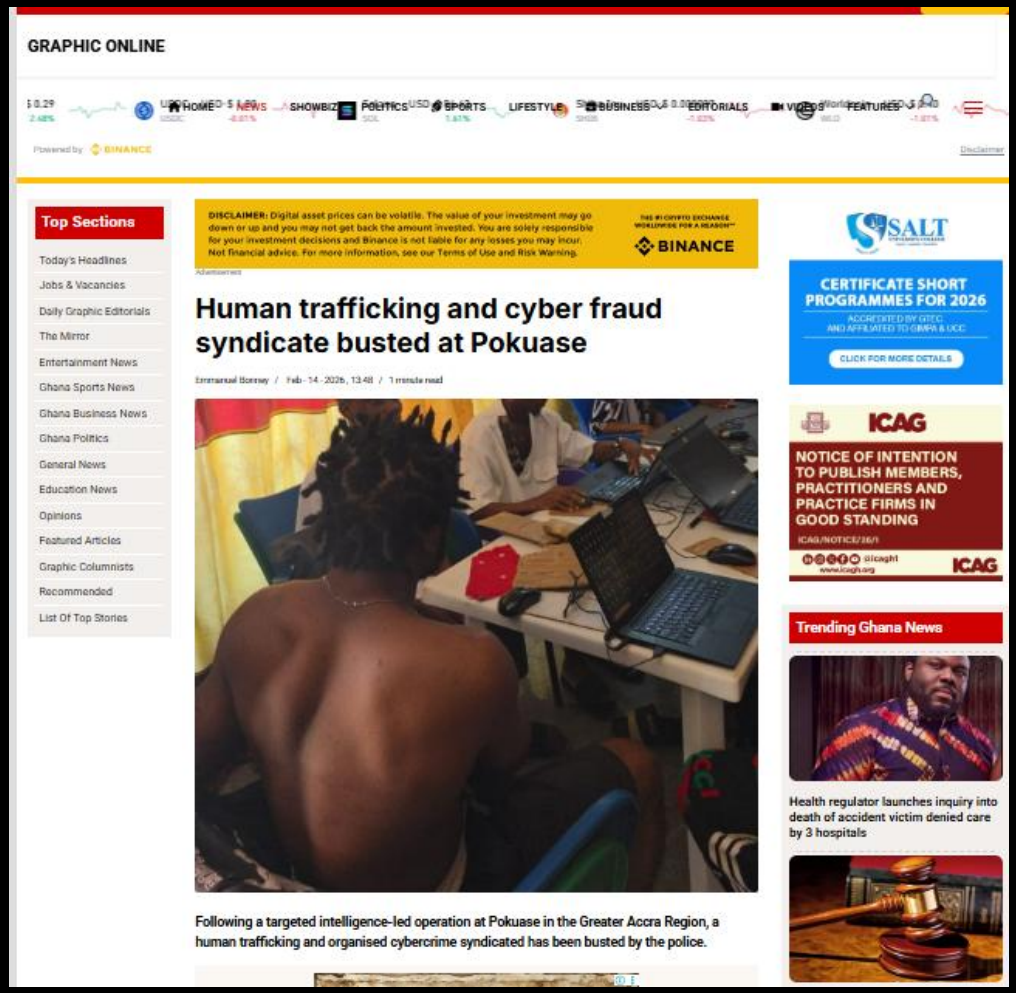
Thomas Naadi, BBC Africa, Accra and Jean Otalor

Cyber Security Authority

The authorities released pictures of some of those found in the raids along with their equipment



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BINANCE

Human trafficking and cyber fraud syndicate busted at Pokuase

Following a targeted intelligence-led operation at Pokuase in the Greater Accra Region, a human trafficking and organised cybercrime syndicated has been busted by the police.

SALT

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NOTICE OF INTENTION TO PUBLISH MEMBERS, PRACTITIONERS AND PRACTICE FIRMS IN GOOD STANDING

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LAB SESSION

LAB SESSION

- Operating Systems & File Management
- https://www.youtube.com/watch?v=R8kXPaw-qRY
- What is Software?
  - What is Systems Software?
    - Types of Systems Software
  - What is Operating Systems?
    - Examples of OS
    - The Purpose of OS
    - The Functions of OS
- File Management
  - Files
  - File types, File names, File extensions, File explorers.
  - Folders/Directories
  - Hierarchical folders
  - File sharing
- Applications Software

LAB SESSION

- File Management
  - is the foundation of **Data Integrity**.
    - If you can't find the data or
    - if it’s named poorly,
    - then it can't become Information.

LAB SESSION

1. Define Your Business Profile

- Discuss with your team and finalize the following details for your startup:
  - **The Industry**: Select a specific sector (e.g., Healthcare, Legal Services, Retail, or Creative Arts).
  - **The Business Type**: Specify the niche (e.g., a modern pharmacy, boutique law firm, or digital marketing agency).
  - **Products/Services**: Identify 3–5 core items or services your business will offer to the market.
  - **The Business Name:** Choose a professional and unique name for your startup.
  - **The Management Structure**: Assign roles to team members (e.g., General Manager, IT Manager, Operations Lead) to ensure participation and accountability throughout the 10-week schedule.

LAB SESSION

2. Digital Organization (Operating System Navigation)

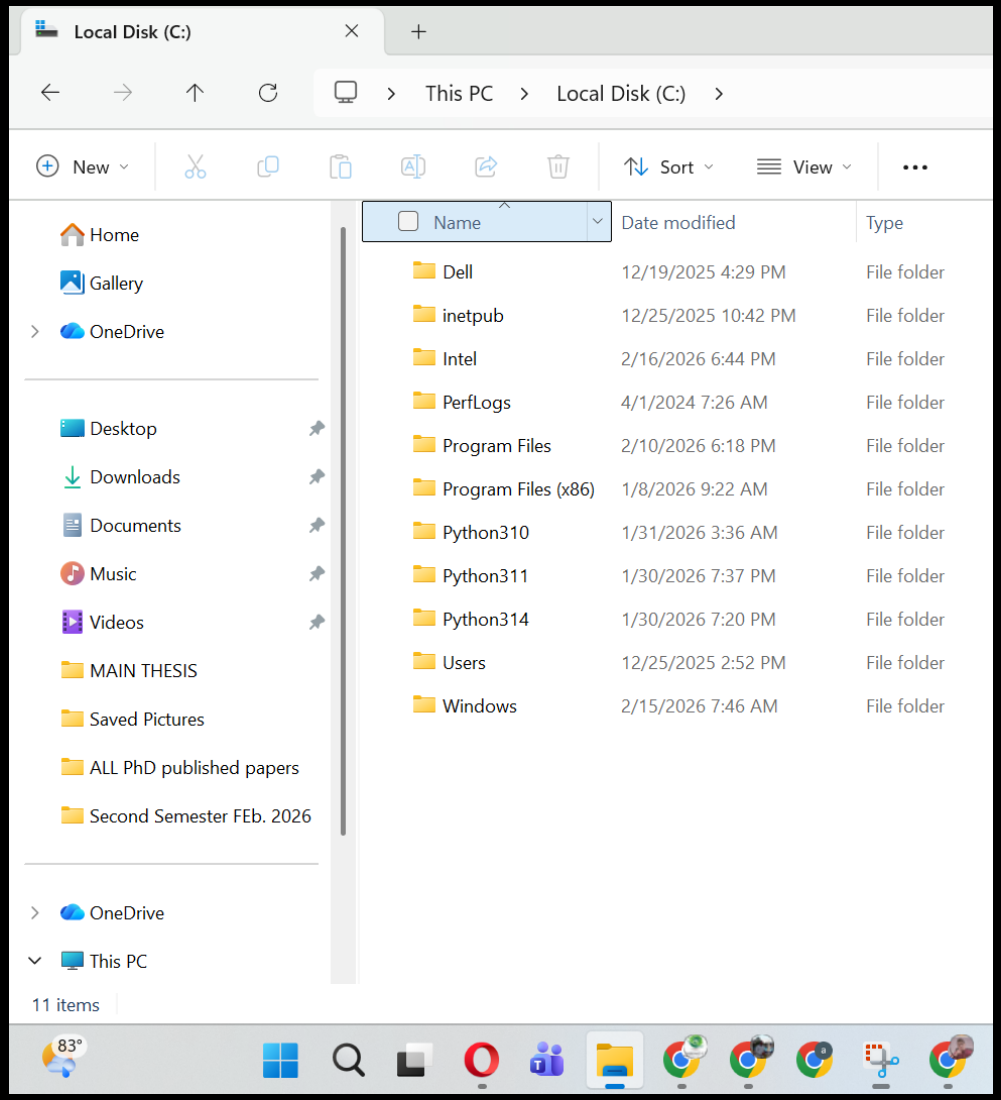
<u>File and Directory/Folder management</u>

Perform the following on your lab workstation:

- **Create a Root Directory**: Create a new folder named exactly after your Business Name.
- **Establish Sub-directories:** Inside your main folder, create the following sub-folders to organize your work for the coming weeks:
  - 01_Documentation (For Word reports)
  - 02_Budget_Analysis (For Excel spreadsheets)
  - 03_Network_Design (For Packet Tracer simulations)
  - 04_Presentations (For the final pitch)

LAB SESSION

- File System Hierarchy
  - C:\
    - ├── Program Files
    - ├── Users
    - └── Documents



LAB SESSION

- Practical Task 1
- Students prepare something similar to:
- Create folder named:

  CSSD111_Enterprise_Project
- Inside it create:
  - 01_Documentation
  - 02_Budget
  - 03_Network
  - 04_Presentation
  - 05_Research

LAB SESSION

- Practical Task 2
- Create a file:
  - Executive_Summary_Draft.docx
- Save it inside 01_Documentation

LAB SESSION

Our Checklist for Week 1

- [ ] Team formed (5–6 members).
- [ ] Business niche selected and approved.
- [ ] Professional folder structure created on the system.
- [ ] One-page "**Business Profile**" draft started in MS Word.

REFLECTION & ANALYSIS QUESTIONS

1. The Literacy vs. Fluency Debate: Why is knowing how to use Microsoft Word or TikTok no longer enough for a business professional in the "Knowledge Age"?
2. DIKW in Action: You receive a notification saying "Server Temperature: 105°C." Identify which part of this is Data, which is Information, and what the resulting Knowledge (action) should be.
3. The Invisible System: Think of a modern ATM. Briefly identify its Hardware, Software, Data, People, and Procedures. What happens to the "System" if the Procedures are missing?
4. Network vs. Web: If a company's internal server goes down, but the building still has an active fiber-optic Internet cable, why can't the employees access their internal files? (Hint: Distinguish between the Network and the Web).
5. Enterprise Impact: How would an Enterprise System (like a shared database) change the way a Marketing team and a Finance team work together compared to each team using their own personal laptops?
6. The Privacy Postcard: We discussed that E-mail is like a "postcard." List three types of business information that you should never send via standard e-mail without encryption.
7. Data Integrity in the Lab: Why does the Lab require a specific numbering system (e.g., 01_Documentation) for your folders? How does this professional organization impact the "Data to Information" pipeline?